FATEMEH GOLSHAN

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EDUCATION

- **Ph.D. in Computer Science.** University of Pittsburgh
  Research Area: Computer Architecture, Computer Systems, Memory Systems. *August 2022 – Present*
  GPA: 3.68/4.0
- **M.S. in Computer Engineering.** Sharif University of Technology
  Major: Computer Architecture *August 2018 – August 2020*
  Thesis: Improving the Accuracy of Data Prefetching via Depth Estimation
  GPA: 18.40/20.0
- **B.S. in Electrical and Computer Engineering.** Isfahan University of Technology
  Major: Hardware Engineering *September 2013 – January 2018*
  Project: Implementation of ARINC 429 Transmitter and Receiver
  GPA: 16.80/20.0

WORK EXPERIENCE

- **Hardware Engineering Intern**, Cisco, San Jose, CA, USA June 2026 – September 2026
  Designing and refining cycle-level simulation frameworks to evaluate next-generation ASIC architectures. Modeling novel multicast modes and advanced congestion test mechanisms for next-generation platforms. Collaborating with silicon design teams to simulate Ethernet Scale-Up Networking (ESUN) workloads, profile high-throughput traffic, and resolve critical bottlenecks in fabric topologies.
- **Software Engineering Intern**, Google, Sunnyvale, CA, USA March 2026 – June 2026
  Engineered an AI-driven on-call assistant leveraging Gemini and Retrieval-Augmented Generation (RAG) to automate root-cause analysis for cluster-level GPU and CPU allocation failures. Built a high-throughput data pipeline to ingest and index large-scale production logs and internal documentation, streamlining incident triage and optimizing infrastructure reliability for massive AI workloads.
- **Hardware Engineering Intern**, Cisco, San Jose, CA, USA May 2025 – August 2025
  Developed large-scale network performance models for the Palladium 2 (PL2) platform, focusing on multicast replication behavior across multi-stage switch fabrics and advanced load-balancing mechanisms. Analyzed hardware performance under high-stress traffic workloads to isolate interconnect bottlenecks and guide future ASIC optimizations.
- **Performance Modeling Consultant**, Qualcomm, Santa Clara, CA, USA September 2024 – December 2024
  Extended advanced memory controller performance and power models across DDR5, LPDDR5, and LPDDR6 standards, focusing on refresh overhead management and scheduling optimizations. Evaluated architectural trade-offs to minimize performance degradation and improve overall system efficiency.
- **Hardware Engineering Intern**, Qualcomm, San Diego, CA, USA May 2024 – August 2024
  Built and evaluated memory controller performance models, explicitly implementing and simulating LPDDR6 Static Efficiency Mode (SEM) to optimize bank resource utilization and mitigate rank-to-rank switching delays. Provided data-driven architectural insights to improve next-generation memory subsystem efficiency.

PUBLICATIONS

- **F. Golshan** and D. Mosse, "CXRay: Full-Visibility CXL Memory Simulation," under review at the IEEE International Symposium on High-Performance Computer Architecture (HPCA), 2027.
- **F. Golshan** and D. Mosse, "Focus on What Matters: DRAM-CXL Hybrid Memory Management with PRISM," International Conference on Parallel Architectures and Compilation Techniques (PACT), 2026.
- M. Bakhshalipour, H. Zare, F. Sarmadi, **F. Golshan**, P. Lotfi-Kamran, and H. Sarbazi-Azad, "Blenda: Dynamically-Reconfigurable Stacked DRAM," *IEEE/ACM International Symposium on Microarchitecture (MICRO)*, 2024.
- A. Ansari, **F. Golshan**, R. Barati, P. Lotfi-Kamran, and H. Sarbazi-Azad, "MANA: Microarchitecting a Temporal Instruction Prefetcher," *IEEE Transactions on Computers (TC)*, 2022.
- M. Shakerinava, **F. Golshan**, A. Ansari, P. Lotfi-Kamran, and H. Sarbazi-Azad, "Evaluation of Data Prefetchers," *Elsevier Journal*, 2022.
- **F. Golshan**, M. Bakhshalipour, M. Shakerinava, A. Ansari, P. Lotfi-Kamran, and H. Sarbazi-Azad, "Harnessing Pairwise-Correlating Data Prefetching with Runahead Metadata," *IEEE Computer Architecture Letters (CAL)*, 2020.
- A. Ansari, **F. Golshan**, P. Lotfi-Kamran, and H. Sarbazi-Azad, "MANA: Microarchitecting an Instruction Prefetcher," *ISCA Instruction Prefetching Championship*, 2020.

SKILLS

- **Programming Languages:** C/C++, Python, CUDA, Verilog, MATLAB
- **Computer Architecture & Systems:** ZSim, Flexus, OMNeT++, Linux `perf`
- **Machine Learning & AI Systems:** PyTorch, Generative AI (Gemini), Retrieval-Augmented Generation (RAG), Graph Neural Networks, Model Optimization
- **Hardware Design & EDA Tools:** ModelSim, Design Compiler, Power Compiler, HSPICE, PSpice, Xilinx ISE
- **Embedded Systems:** ARM, AVR, Raspberry Pi, Arduino
- **Tools:** Git, Bash, LaTeX